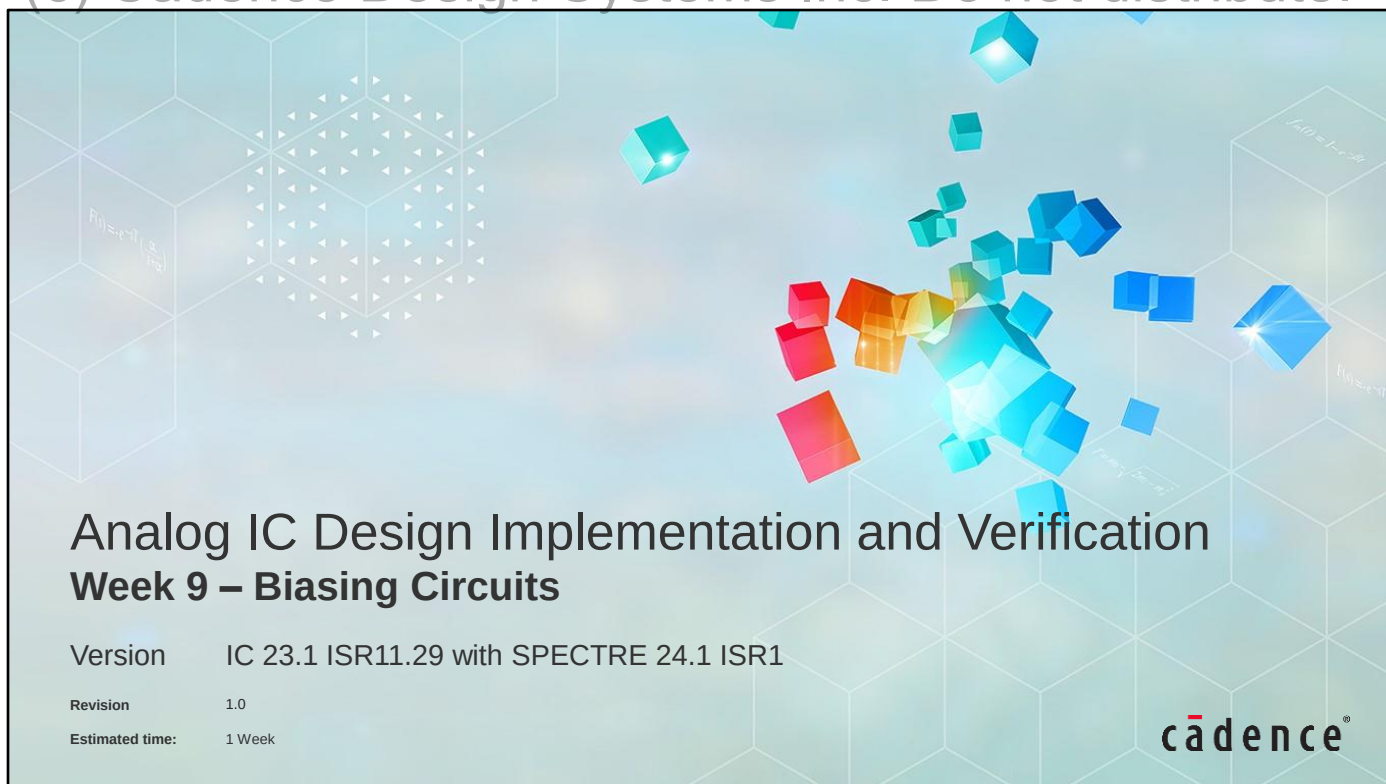




Analog IC Design Implementation and Verification
Week 9 – Biasing Circuits

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Module Objectives

In this module, you

- Trace the development of bandgap voltage references
- Characterize and model current references
- Design bias circuits to reduce the effects of noise and interference



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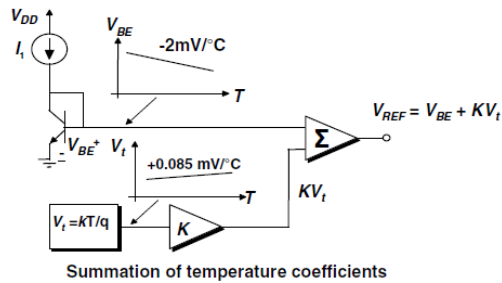
Temperature Stable References

The bandgap voltage concept combined with power supply independence creates a very stable voltage reference in regard to both temperature and power supply variations.

Bandgap voltage reference principle: To balance the negative temperature coefficient (TC_F) of a pn junction with the positive temperature coefficient of the thermal voltage $V_t = kT/q$.

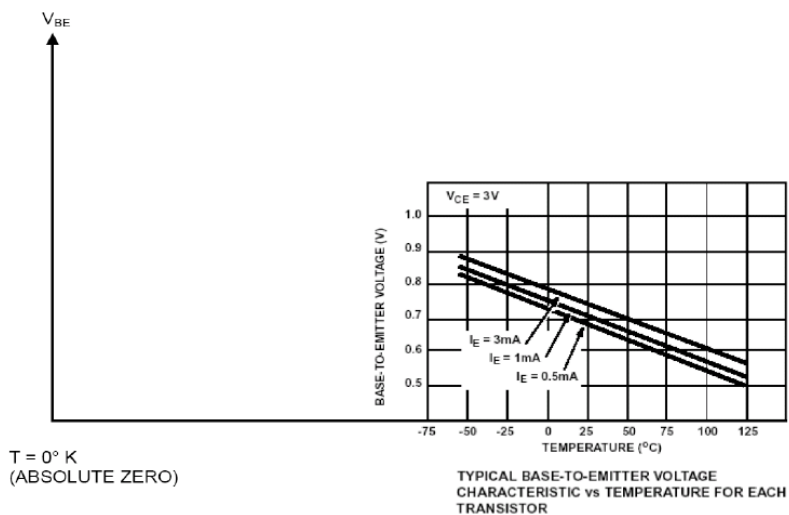
Concept: Summation of counter-balancing temperature coefficients.

Result: References with TC_F s approaching 10 ppm/°C.



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Bandgap Principle: V_{BE} vs. Temperature



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Temperature Coefficient of Base-Emitter Voltage

Follow these steps to derive the temperature coefficient of the base-emitter voltage. (For small TCFs the dependence V_{BE} must be known more precisely than $\approx -2 \text{ mV}/^\circ\text{C}$.)

1. Start with the collector current density, J_C : $J_C = \frac{q \bar{D}_n n_{pe}}{W_B} \exp\left(\frac{V_{BE}}{V_t}\right)$

where, J_C = I_C/Area = collector current density

$\bar{D}_n$ = average diffusion constant for electrons

W_B = base width

V_{BE} = base-emitter voltage

V_t = kT/q

k = Boltzmann's constant ($1.38 \times 10^{-23} \text{ J}/^\circ\text{K}$)

T = Absolute temperature

n_{po} = n_i^2/N_A = equilibrium concentration of electrons in the base

$$n_i^2 = DT^3 \exp\left(\frac{-V_{GO}}{V_t}\right) = \text{intrinsic concentration of carriers}$$

D = temperature independent constant

V_{GO} = bandgap voltage of silicon (1.205V)

N_A = acceptor impurity concentration



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Deriving Temperature Coefficient (continued)

2. Combine the above relationships into one:

$$J_C = \frac{q\bar{D}}{N_A W_B} D T^3 \exp\left(\frac{V_{BE} - V_{GO}}{V_i}\right) = A T^\gamma \exp\left(\frac{V_{BE} - V_{GO}}{V_i}\right) \quad \text{where, } \gamma = 3$$

3. The value of J_C at a reference temperature of $T = T_0$ is

$$J_{C0} = A T_0^\gamma \exp\left[\frac{q}{k T_0} (V_{BE} - V_{GO})\right]$$

while the value of J_C at a general temperature T is

$$J_C = A T^\gamma \exp\left[\frac{q}{k T} (V_{BE} - V_{GO})\right]$$

4. The ratio of J_C/J_{C0} can be expressed as $\underline{J_C}$

$$\frac{J_C}{J_{C0}} = \left(\frac{T}{T_0}\right)^\gamma \exp\left[\frac{q}{k} \left(\frac{V_{BE} - V_{GO}}{T} - \frac{V_{BE0} - V_{GO}}{T_0}\right)\right]$$

or

$$\ln\left(\frac{J_C}{J_{C0}}\right) = \gamma \ln\left(\frac{T}{T_0}\right) + \frac{q}{k T} \left[V_{BE} - V_{GO} - \frac{T}{T_0} (V_{BE0} - V_{GO})\right]$$

where V_{BE0} is the value of V_{BE} at $T = T_0$



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Deriving Temperature Coefficient (continued)

5. Solving for V_{BE} from the previous results gives

$$V_{BE}(T) = V_{\infty} \left(1 - \frac{T}{T_0} \right) + V_{BE} \left(\frac{T}{T_0} \right) + \frac{\gamma k T}{q} \ln \left(\frac{T_0}{T} \right) + \frac{k T}{q} \ln \left(\frac{J_C}{J_{C0}} \right)$$

6. Next, assume $J_C \propto T^\alpha$ and find

$$\frac{\partial V_{BE}}{\partial T} = \frac{\partial V_{GO}}{\partial T} \left(1 - \frac{T}{T_0} \right) + \frac{V_{GO} + V_{BE0}}{T_0} + \frac{\gamma k T}{q} \frac{\partial \ln(T_0/T)}{\partial T} + \ln \left(\frac{T_0}{T} \right) \frac{\partial (\gamma k T/q)}{\partial T} + \frac{k T}{q} \left(\frac{\partial \ln(J_C/J_{C0})}{\partial T} \right) + \left(\frac{k}{q} \right) \ln \left(\frac{J_C}{J_{C0}} \right)$$

7. Assume that $T = T_0$, which means $J_C = J_{C0}$. Because, $\frac{\partial V_{BE}}{\partial T} = 0$

$$\frac{\partial V_{BE}}{\partial T} \Big|_{T=T_0} = -\frac{V_{GO}}{T_0} + \frac{V_{BE0}}{T_0} + \frac{\gamma k T}{q} \cdot \frac{\partial \ln(T_0/T)}{\partial T} + \frac{k T}{q} \left(\frac{\partial \ln(J_C/J_{C0})}{\partial T} \right)$$



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Deriving Temperature Coefficient (continued)

8. Notice that

$$\frac{\partial \ln(I_0/T)}{\partial T} = \frac{T}{T_0} \frac{\partial (I_0/T)}{\partial T} = \frac{T}{T_0} \left(\frac{-1}{T^2} \right) = -\frac{1}{T} \quad \text{and} \quad \frac{\partial \ln(J_C/J_{C0})}{\partial T} = \frac{J_{C0}}{J_C} \frac{\partial (J_C/J_{C0})}{\partial T} = \frac{J_{C0}}{J_C} \left(\frac{\alpha J_C}{T J_{C0}} \right) = \frac{\alpha}{T}$$

Therefore,

$$\frac{\partial V_{BE}}{\partial T} \bigg|_{T=T_0} = -\frac{V_{GQ}}{T_0} + \frac{V_{BE0}}{T_0} - \frac{\gamma k}{q} + \frac{k}{q} \quad \text{or} \quad \frac{\partial V_{BE}}{\partial T} \bigg|_{T=T_0} = \frac{V_{BE0} - V_{GQ}}{T_0} + (\alpha - \gamma) \left(\frac{k}{q} \right) \bigg|$$

Typical values of α and γ are 1 and 3.2. If $V_{BE0} = 0.6\text{V}$, then at room temperature:

$$\frac{\partial V_{BE}}{\partial T} \bigg|_{T=T_0} = \frac{0.6 - 1.205}{300} + (1 - 3.2) \left(\frac{0.026}{300} \right) = \frac{0.6 - 1.205 - 0.1092}{300} = -1.826 \text{ mV}/^\circ\text{C}$$

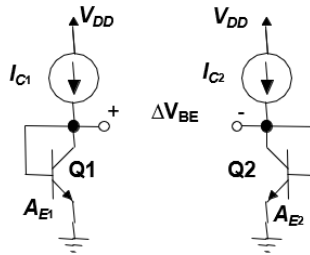


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Deriving Thermal Voltage Temperature Coefficient

Follow these steps to derive the thermal voltage temperature coefficient (kT/q):

- Consider two identical pn junctions having different current densities:



$$\Delta V_{BE} = \Delta V_{BE1} - \Delta V_{BE2} = \frac{kT}{q} \ln \left(\frac{J_{C1}}{J_{C2}} \right)$$

$$\partial(\Delta V_{BE}) / \partial T,$$

- Find $\frac{\partial(\Delta V_{BE})}{\partial T} = \frac{V_T}{T} \ln \left(\frac{J_{C1}}{J_{C2}} \right) = \frac{k}{q} \ln \left(\frac{J_{C1}}{J_{C2}} \right)$

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Deriving Gain and K: Bandgap Voltage Reference

1. To achieve a zero temperature coefficient at $T = T_0$, the following equation must be satisfied:

$$0 = \frac{\partial(V_{BE})}{\partial T} \bigg|_{T=T_0} + K'' \frac{\partial(\Delta V_{BE})}{\partial T} \quad \text{where } K'' \text{ is a constant that satisfies the equation.}$$

2. Therefore, you get

$$0 = K'' \left(\frac{V_{BE}}{T_0} \right) \ln \left(\frac{J_{C1}}{J_{C2}} \right) + \frac{V_{BE} - V_{BE0}}{T_0} + \frac{(\alpha - \gamma)V_{BE0}}{T_0}$$

3. Define $K = K'' \ln \left(\frac{J_{C1}}{J_{C2}} \right)$ therefore,

$$0 = K'' \left(\frac{V_{BE}}{T_0} \right) + \frac{V_{BE} - V_{BE0}}{T_0} + \frac{(\alpha - \gamma)V_{BE0}}{T_0}$$

4. Solving for K gives $K = \frac{V_{BE0} - V_{BE0}}{V_{BE0}} \frac{V_{BE0}}{V_{BE0}} \frac{V_{BE0}}{V_{BE0}} (\alpha - \gamma)$

Assuming that $J_{C1}/J_{C2} = A_{E1}/A_{E2} = 10$ and $V_{BE0} = 0.6V$ gives

$$K = \frac{1.205 - 0.6}{0.026} (2.2)(0.026) = 25.469$$

5. The output voltage of the bandgap voltage reference is found as

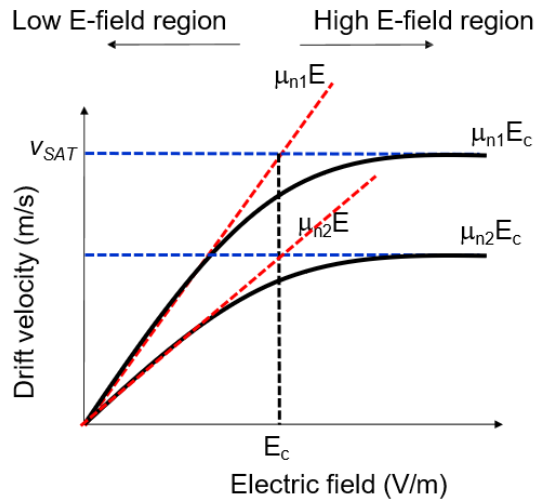
$$V_{REF,T=T_0} = V_{BE0} + K V_{BE0} = V_{BE0} + V_{BE0} (\alpha - \gamma) V_{BE0}$$

$$\text{For the previous values, } V_{REF} = V_{BE0} + (\alpha - \gamma) V_{BE0}$$



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Negative TC Voltages



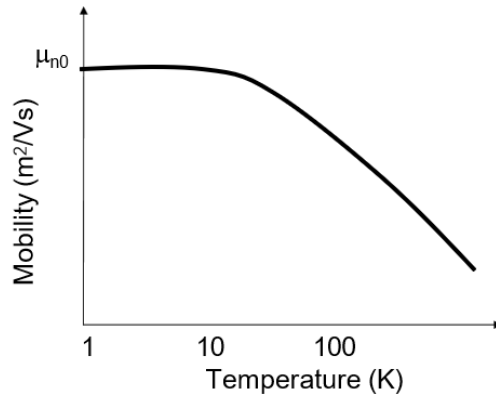
$$v_{drift}(T) = \frac{\mu_n(T)E}{1 + \frac{E}{E_c}}$$

$$\mu_n(T) \cong \mu_{n0} T^{-3/2}$$



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Negative TC Voltages (continued)

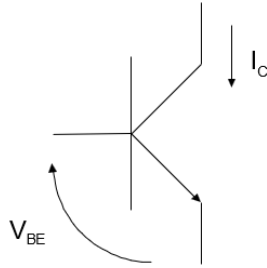


$$\mu_n(T) \cong \mu_{n0} T^{-3/2}$$

$$\frac{d\mu_n(T)}{dT} \cong -\frac{3}{2} \mu_{n0} T^{-5/2}$$



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Negative TC Voltages (continued)

$$I_C = I_S \left(e^{\frac{qV_{BE}}{kT}} - 1 \right)$$

$$V_{BE} = \frac{kT}{q} \ln \left(\frac{I_C}{I_S} \right) \Rightarrow I_S \propto \mu_n(T)$$

$$I_S \cong pT^{2.5} e^{-\frac{E_{gap}}{kT}}$$

$$\frac{\partial V_{BE}}{\partial T} = \frac{k}{q} \ln \left(\frac{I_C}{I_S} \right) - \frac{kT}{q} \left(\frac{1}{I_S} \frac{\partial I_S}{\partial T} - \frac{1}{I_C} \frac{\partial I_C}{\partial T} \right)$$

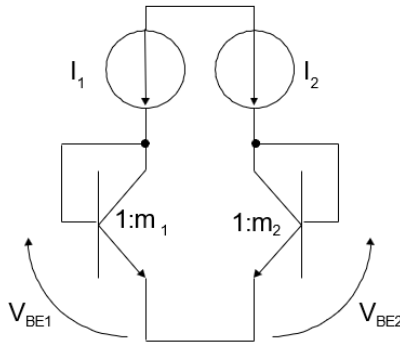
Temperature coefficient $\rightarrow \frac{\partial V_{BE}}{\partial T} < 0$

$$\frac{\partial(\ln u)}{\partial x} = u \frac{\partial v}{\partial x} + v \frac{\partial u}{\partial x}$$

$$\frac{\partial(\ln u)}{\partial x} = \frac{1}{u} \frac{\partial u}{\partial x}$$

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Positive TC Voltages



$$V_{BE1} = \frac{kT}{q} \ln \left(\frac{I_1}{I_{S1}} \right)$$

$$V_{BE2} = \frac{kT}{q} \ln \left(\frac{I_2}{I_{S2}} \right)$$

$$\Delta V_{BE} = V_{BE1} - V_{BE2} = \frac{kT}{q} \ln \left(\frac{I_{S2} I_1}{I_{S1} I_2} \right)$$

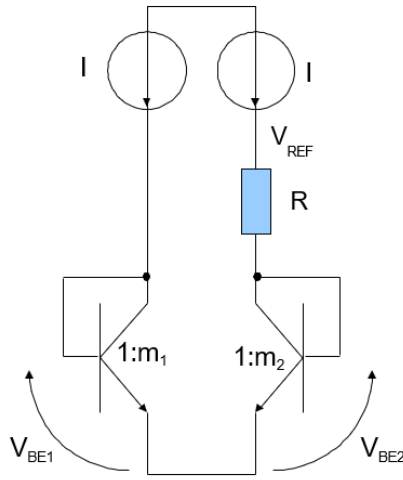
$$\text{For } \frac{I_{S2}}{I_{S1}} = \frac{m_2}{m_1} = m \text{ and } \frac{I_1}{I_2} = n$$

$$\Delta V_{BE} = \frac{kT}{q} \ln(mn)$$

$$\text{Temperature coefficient} \rightarrow \frac{\partial \Delta V_{BE}}{\partial T} = \frac{k}{q} \ln(mn) > 0$$

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Zero TC Voltage (Bandgap Reference)



$$V_{BE1} = \frac{kT}{q} \ln\left(\frac{I}{I_{S1}}\right)$$

$$V_{BE2} = \frac{kT}{q} \ln\left(\frac{I}{I_{S2}}\right)$$

$$RI = \Delta V_{BE} = \frac{kT}{q} \ln(m)$$

$$V_{REF} = V_{BE2} + RI$$

$$V_{REF} = V_{BE2} + \frac{kT}{q} \ln(m)$$

$$\frac{\partial V_{REF}}{\partial T} = \frac{\partial V_{BE2}}{\partial T} + \frac{k}{q} \ln(m) \cong 0$$

Negative dependence

Positive dependence

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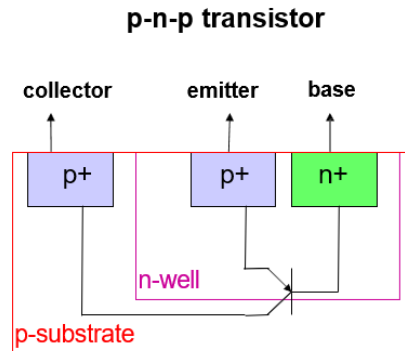
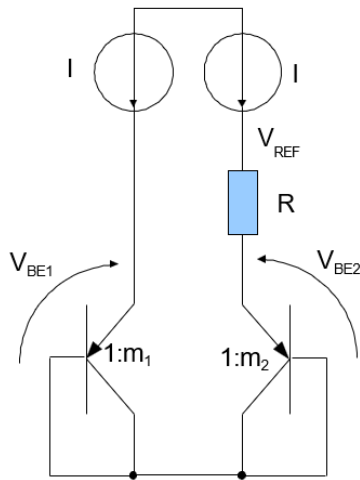
Bandgap Reference Sources

- You need to guarantee that V_{REF} equals V_{BE1}
- To guarantee the condition $V_{REF} = V_{BE1}$, a common-mode feedback network must be added
- For CMOS process, a p-n-p transistor rather than a n-p-n is used



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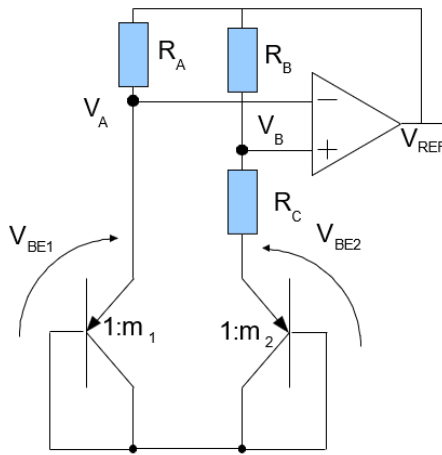
Zero TC Voltage (Bandgap Reference)



$$V_{REF} = V_{BE2} + \frac{kT}{q} \ln \left(\frac{m_2}{m_1} \right)$$

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Bandgap Reference



$$m = \frac{m_2}{m_1}$$

$$V_A = V_B = V_{BE2} \quad R_C I$$

$$I = \frac{kT}{R_C q} \ln(m)$$

$$V_{REF} = V_B + IR_B$$

$$V_{REF} = V_{BE2} + \frac{kT}{q} \ln(m) \left[1 + \frac{R_B}{R_C} \right]$$

$$\frac{\partial V_{REF}}{\partial T} = \frac{\partial V_{BE2}}{\partial T} + \frac{k}{q} \ln(m) \left[1 + \frac{R_B}{R_C} \right]$$

Negative dependence

Positive dependence



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Bandgap: Hand Calculations

$$\frac{\partial V_{BE}}{\partial T} = TC_{VT} + TC_{IS} + TC_{IC}$$

$$\frac{\partial V_{BE}}{\partial T} = \frac{\partial}{\partial T} \left[\frac{kT}{q} \ln \left(\frac{I_C}{I_S} \right) \right] = \frac{kT}{qI_S} \frac{\partial I_S}{\partial T} + \frac{kT}{qI_C} \frac{\partial I_C}{\partial T}$$

• TC_{VT}

$$\frac{\partial V_T}{\partial T} = \frac{\partial (kT/q)}{\partial T} = \frac{k}{q} \Rightarrow TC_{VT} = \frac{k}{q} \ln \left(\frac{I_C}{I_S} \right) \cong 0.085 \ln \left(\frac{I_C}{I_S} \right) \frac{mV}{^{\circ}C}$$

• TC_{IS}

$$TC_{IS} = \frac{kT}{qI_S} \frac{\partial I_S}{\partial T} \cong 2.5 \frac{k}{q} + \frac{E_G}{q} = 0.2125 + \frac{E_G}{qT} \frac{mV}{^{\circ}C}$$

• TC_{IC}

$$\frac{\partial I_C}{\partial T} \cong \frac{I_C}{T} \Rightarrow TC_{IC} = V_T \frac{I_C}{T} = \frac{k}{q} \cong 0.085 \frac{mV}{^{\circ}C}$$



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Hand Calculations (continued)

$$\frac{\partial V_{BE}}{\partial T} = TC_{VT} + TC_{IS} + TC_{IC}$$

$$\frac{\partial V_{BE}}{\partial T} = 0.085 \ln \left(\left| \frac{I_C}{I_S} \right| \right) - 0.2125 - \frac{E_G}{q} + 0.085 \frac{mV}{^\circ C}$$

$$\frac{\partial V_{BE}}{\partial T} = 0.085 \ln \left(\left| \frac{I_C}{I_S} \right| \right) - \left(0.1275 + \frac{E_G}{qT} \right) \frac{mV}{^\circ K}$$

$$\frac{\partial V_{BE}}{\partial T} = 0.6225 - \frac{E_G}{qT} \frac{mV}{^\circ K}$$

$$\frac{\partial V_{BE}}{\partial T} = -1.585 \frac{mV}{^\circ K}$$



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Hand Calculations (continued)

Case 1

$$\frac{\partial V_{REF}}{\partial T} = \frac{\partial V_{BE2}}{\partial T} + \frac{k}{q} \ln(m) = 0$$

$$\frac{k}{q} \ln(m) = 1.585 \frac{mV}{^{\circ}K}$$

$$\ln(m) = 18.64$$

$$m = e^{18.64} = 125 \times 10^6$$

The ratio m is prohibitive large.

Case 2

$$\frac{\partial V_{REF}}{\partial T} = \frac{\partial V_{BE2}}{\partial T} + \frac{k}{q} \ln(m) \left[1 + \frac{R_s}{R_c} \right] = 0$$

$$\frac{k}{q} \ln(m) \left[1 + \frac{R_s}{R_c} \right] = 1.585 \frac{mV}{^{\circ}K}$$

$$\ln(m) \left[1 + \frac{R_s}{R_c} \right] = 18.64$$

$$m = e^{18.64 R_c / (R_s + R_c)}$$

$$\text{For } R_s \cong 8R_c$$

$$m \cong e^{2.07} \cong 8$$

The ratio m is reasonable.



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Hand Calculations (continued)

Case 2 (continued)

- Device ratio: 1:m = 1:8
- Current

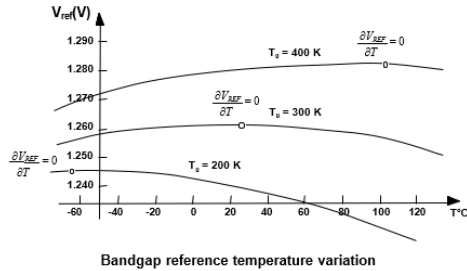
$$I = \frac{kT}{R_c q} \ln(m) \cong \frac{0.0258}{1245} \ln(8) \cong 44 \mu A$$



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Variation: Bandgap Reference Voltage/Temperature

The previous derivation is valid only at the given temperature T_0 . Solving for the zero temperature coefficient at different temperatures produces different curves.



Bandgap curvature correction is necessary for low ppm/C bandgap references.

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Classical Widlar Bandgap Voltage Reference

Operation

$$V_{BE1} = V_{BE2} \quad I_2 R_3 \quad \text{gives } \Delta V_{BE} = V_{BE1} \quad V_{BE2} = I_2 R_3$$

$$\text{But } \Delta V_{BE} = V_T \ln \left(\frac{I_1}{I_2} \right) - V_T \ln \left(\frac{I_2}{I_1} \right) = V_T \ln \left(\frac{I_1 I_2}{I_2 I_1} \right)$$

$$\text{Assume } V_{BE1} \approx V_{BE3}, \text{ we get } I_1 R_1 = I_2 R_2$$

Therefore,

$$I_2 = \frac{\Delta V_{BE}}{R_3} = \frac{V_T}{R_3} \ln \left(\frac{I_1 I_2}{I_2 I_1} \right) = \frac{V_T}{R_3} \ln \left(\frac{R_2 I_2}{R_1 I_1} \right)$$

Now we can express V_{REF} as

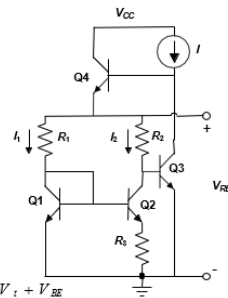
$$V_{REF} = I_2 R_2 + V_{BE3} = \frac{R_2}{R_3} V_T \ln \left(\frac{R_2 I_2}{R_1 I_1} \right) + V_{BE3} = K V_T + V_{BE}$$

Design R_1 , R_2 , I_{s1} , and I_{s2} to get the desired K .

Let $K = 25$ and $I_{s2} = 10 I_{s1}$ and design R_1 , R_2 , and R_3 . Choose $R_2 = 10 R_1 = 10 \text{ k}\Omega$.

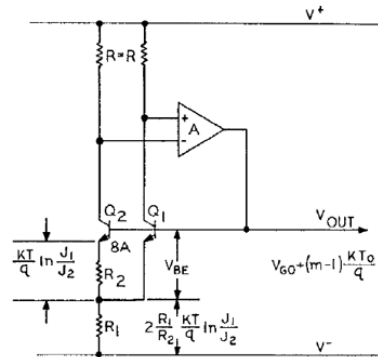
Therefore, $\ln(100) = 4.602$. Therefore $R_2/R_3 = 25/4.602$ or $R_3 = R_2/5.4287 = 1.842 \text{ k}\Omega$.

R.J. Widlar, "New Developments in IC Voltage Regulators," IEEE J. of Solid-State Circuits, Vol. SC-8, pp. 2-7, February 1971. ECE 4430 —Analog Integrated Circuits and Systems



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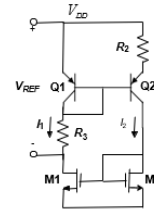
Bandgap Principle: Brokaw Bandgap Circuit



Idealized circuit showing two-transistor bandgap cell

Reference: A. P. Brokaw, "A simple 3-terminal IC bandgap voltage reference," *IEEE JSSC*, Dec., 1974
BA01: Analog IC Design, Part 2

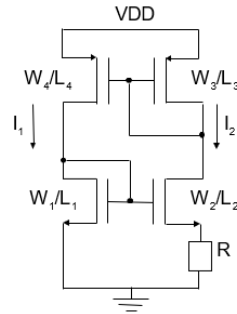
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The Supply-Independent Reference Source

$$\begin{aligned}
 V_{GS1} &= V_{GS2} = I_2 R \\
 I_1 &= \frac{\beta_1}{2} (V_{GS1} - V_{T1})^2 \therefore V_{GS1} = \sqrt{\frac{2I_1}{\beta_1}} + V_{T1} \\
 \sqrt{\frac{2I_1}{\beta_1}} + V_T &= \sqrt{\frac{2I_2}{\beta_2}} + V_T + I_2 R \\
 \sqrt{\frac{2I_1}{\beta_1}} - \sqrt{\frac{2I_2}{\beta_2}} &= I_2 R \\
 I_2 = I_1 = I \\
 \sqrt{\frac{2I}{\beta_1}} \left(1 - \sqrt{\frac{W_1 L_2}{W_2 L_1}} \right) &= IR \\
 I = \frac{2}{R^2 \beta_1} \left(1 - \sqrt{\frac{W_1 L_2}{W_2 L_1}} \right)^2 &\Rightarrow \frac{\partial I}{\partial V_{DD}} = 0
 \end{aligned}$$



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CMOS Bandgap Reference: Substrate PNP BJTs

Operation

The cascode mirror (M5-M8) keeps the currents in Q1, Q2, and Q3 identical.

Thus,

$$V_{BE1} = I_2 R \quad V_{BE2}$$

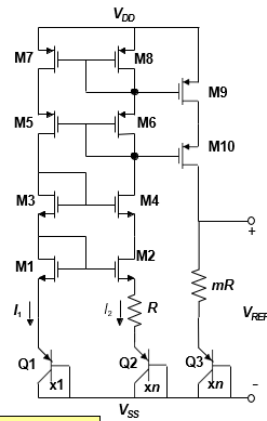
or

$$I_2 = \frac{V_t}{R} \ln(n)$$

Therefore,

$$V_{REF} = V_{BE3} \quad I_2(mR) = V_{BE3} \quad mV_t \ln(n)$$

Use m and n to design the desired value of K (n is an integer greater than 1).



Note: Most circuits use low-voltage cascodes to enable lower V_{DD} .

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Weak Inversion Bandgap Voltage Reference

For the p-channel transistors

$$I_D = I_{D0}(W/L)\exp\left(\frac{V_{BG}}{nV_t}\right)\left[\exp\left(\frac{-V_{BS}}{V_t}\right) - \exp\left(\frac{-V_{BD}}{V_t}\right)\right]$$

where $V_t = kT/q$

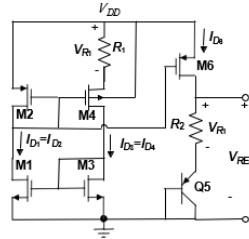
If $V_{BD} \gg V_t$, then $I_D = I_{D0}(W/L)\exp\left(\frac{V_{BG}}{nV_t} - \frac{V_{BS}}{V_t}\right)$

The various transistor currents can be expressed as

$$I_{D1} = I_{D2} = I_{D0}(W_1/L_1)\exp\left(\frac{V_{BG1}}{nV_t}\right) \quad \text{and} \quad I_{D3} = I_{D4} = I_{D0}(W_4/L_4)\exp\left(\frac{V_{BG4}}{nV_t} - \frac{V_{BS4}}{V_t}\right)$$

Notice that $V_{BG2} = V_{BG4}$ and $V_{BS4} = V_{R1}$

$$\therefore \frac{I_{D1}}{I_{D3}} = \frac{W_1/L_1}{W_4/L_4}\exp\left(\frac{V_{R1}}{V_t}\right) \quad \text{where} \quad V_{R1} = V_t \ln\left(\frac{W_1W_3L_2L_3}{L_1L_4W_2W_5}\right) \quad \text{and} \quad I_{R1} = \frac{V_{R1}}{R_1}$$



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Weak Inversion Bandgap Voltage Reference (continued)

The reference voltage can be expressed as

$$V_{REF} = R_2 I_6 + V_{BE5}$$

$$\text{However, } I_6 = \frac{W_6 L_3}{L_6 W_3} I_{B1} = \frac{W_6 L_3}{L_6 W_3} \frac{V_i}{R_1} \ln \left(\frac{W_1 W_4 L_2 L_3}{L_1 L_4 W_2 W_3} \right)$$

Substituting I_6 and the previously derived expression for $V_{BE}(T)$ in V_{REF} gives

$$V_{REF} = \frac{W_6 L_3}{L_6 W_3} \frac{R_2}{R_1} V_i \ln \left(\frac{W_1 W_4 L_2 L_3}{L_1 L_4 W_2 W_3} \right) + V_{\infty} \left(1 - \frac{T}{T_0} \right) + V_{BE0} \left(\frac{T}{T_0} \right) + 3 V_i \ln \left(\frac{T_0}{T} \right)$$

To achieve $\frac{\partial V_{REF}}{\partial T} = 0$ at $T = T_0$, we get

$$\frac{\partial V_{REF}}{\partial T} = \left(\frac{k}{q} \right) \left(\frac{R_2}{R_1} \right) \left(\frac{W_6 L_3}{L_6 W_3} \right) \ln \left(\frac{W_1 W_4 L_2 L_3}{L_1 L_4 W_2 W_3} \right) - \frac{V_{\infty}}{T_0} + \frac{V_{BE0}}{T_0} + \frac{3k}{q}$$

$$\text{Therefore } \frac{R_2 W_6 L_3}{R_1 L_6 W_3} \ln \left(\frac{W_1 W_4 L_2 L_3}{L_1 L_4 W_2 W_3} \right) = \frac{q}{k T_0} (V_{\infty} - V_{BE0}) - 3$$

Under the above constraint, V_{REF} has an $\approx$ zero TC_F at $T = T_0$ and has a value of

$$V_{REF} = V_{\infty} + \frac{3kT}{q} \left[1 + \ln \left(\frac{T_0}{T} \right) \right] = V_{\infty} + \frac{3kT}{q}$$

Practical values of $\frac{\partial V_{REF}}{\partial T}$ for the weak inversion bandgap are less than 100 ppm/°C.



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Temperature- and Supply-Independent Biasing

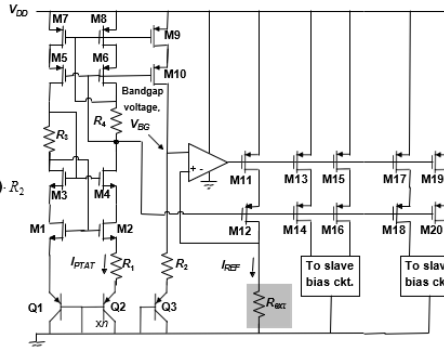
A temperature- and supply-independent current source and its distribution are shown.

Constant current:

$$I_{REF} = \frac{V_{BG}}{R_{ext}}$$

where

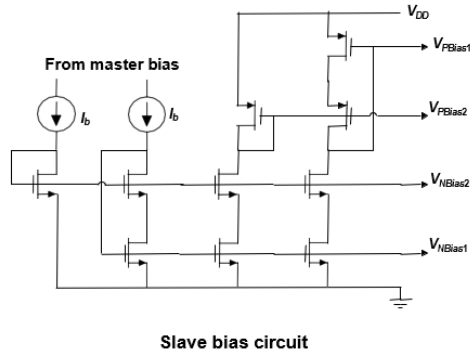
$$V_{BG} = V_{BE3} + I_{PTAT}R_2 = V_{BE3} + \frac{V_T}{R_1} \ln(n) \cdot R_2$$



Note: R_{ext} is often external to the chip to get a tighter tolerance on the I_{REF} current.

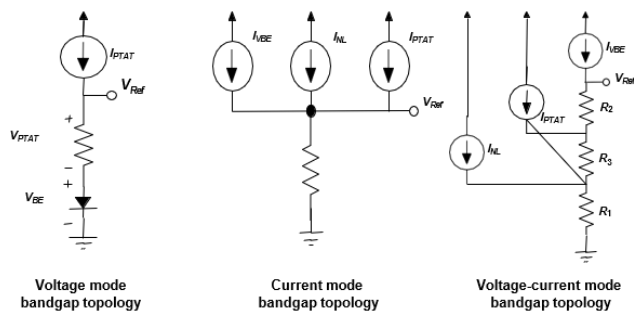
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Biasing (continued)



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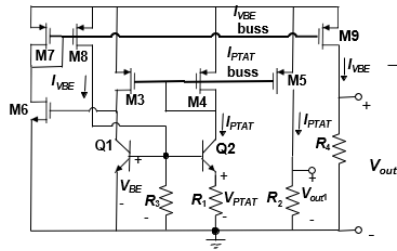
Bandgap Topologies: Low-Voltage Power Supply



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Generating Currents with Temperature Coefficients

Generating currents with VBE and PTAT temperature coefficients

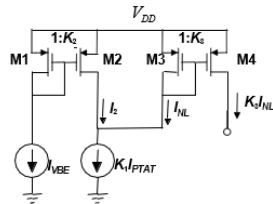


$$V_{OUT1} = I_{PTAT} R_2 = \left(\frac{V_{PTAT}}{R_1} \right) R_2 = V_{PTAT} \frac{R_2}{R_1}$$

$$V_{OUT2} = I_{VBE} R_4 = \left(\frac{V_{BE}}{R_3} \right) R_4 = V_{BE} \frac{R_4}{R_3}$$

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Canceling Bandgap Curvature



Circuit to generate nonlinear correction term, I_{NL}

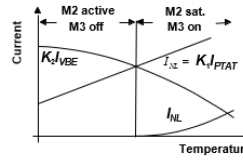


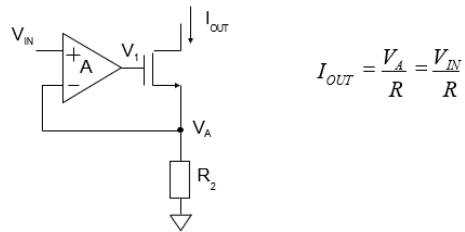
Illustration of various currents

$$I_{NL} = \begin{cases} 0, & K_2 I_{VBE} > K_1 I_{PTAT} \\ K_1 I_{PTAT} - K_2 I_{VBE}, & K_2 I_{VBE} < K_1 I_{PTAT} \end{cases}$$

The combination of this concept with the temperature coefficients from the previous slide yields a curvature-corrected bandgap reference of 0.596V with a TC of 20 ppm/°C from -15°C to 90°C using a 1.1V power supply. In addition, the line regulation is 408 ppm/V for $1.2 \leq V_{DD} \leq 10V$ and 2000 ppm/V for $1.1 \leq V_{DD} \leq 10V$. Quiescent current is 14 μA .

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Alternative ZTAT Current Source



Temperature independent supply voltage: $V_{IN} = V_{BANDGAP}$

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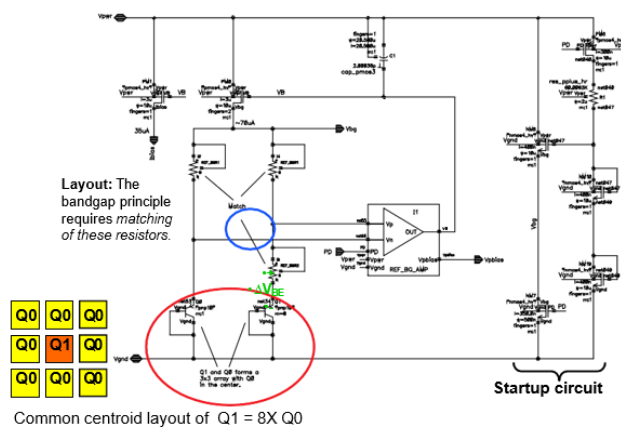
Startup Circuit

- Bandgap references have multiple stability points, necessitating the addition of a startup circuit
- Two functions of the startup circuit:
 - Remove the circuit from undesired mode
 - Do not affect operation if you are in the desired mode



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Bandgap Voltage Reference Block



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Practical Aspects

Correction of second-order effects in the bandgap voltage reference can achieve very stable (1 ppm/°C) voltage references

- Watch out for second-order effects, such as noise, when using the bandgap voltage reference in sensitive applications
- Bandgap voltage references can achieve temperature dependence less than 50 ppm/°C
- Bandgap references have multiple stability points, necessitating the addition of a startup circuit
- Bandgap voltage references are independent of the power supply
- Need correction circuitry to achieve smaller values of temperature coefficient
- Real bandgap needs some frequency compensation like an opamp
- Frequency compensation usually reduces PSRR at higher frequencies
- Frequency compensation usually reduces high-frequency noise



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Introduction: Current References

- Simple current references
- Widlar current references
- Peaking current references
- Threshold-referenced references
- The concept of sensitivity



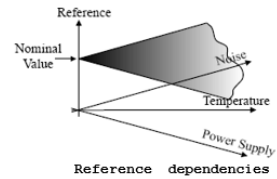
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What Is Reference Dependence?

Voltage or current reference: An independent voltage or current source that has a high degree of precision and stability

Requirements of a reference circuit should be independent of:

- Power supply
- Temperature
- Processing variations
- Noise and other interference



Note: Some references have a desired temperature dependence to compensate for other effects. Also, instead of building a current reference, you can build a voltage reference and then use a voltage-to-current converter.



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What Is Power Supply Independence?

Use the concept of sensitivity:
$$S_{V_{DD}}^{I_{REF}} = \frac{\partial I_{REF} / I_{REF}}{\partial V_{DD} / V_{DD}} = \frac{V_{DD}}{I_{REF}} \left(\frac{\partial I_{REF}}{\partial V_{DD}} \right)$$

Apply sensitivity to determine power supply dependence:

$$\frac{\partial I_{REF}}{I_{REF}} = \left(S_{V_{DD}}^{I_{REF}} \right) \frac{\partial V_{DD}}{V_{DD}}$$

Thus, the fractional change in the reference voltage is equal to the sensitivity times the fractional change in the power supply voltage. For example, if the sensitivity is 1, then a 10% change in V_{DD} causes a 10% change in I_{REF} .

Ideally, you want $S_{V_{DD}}^{I_{REF}}$ to be zero for power supply independence.

Note: Often $20 \cdot \log|S|$ is used to get more intuitive values in dB.

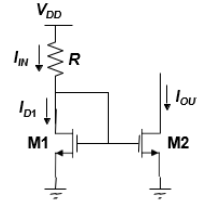


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Simple Current Reference

$$I_{OUT} \approx \frac{V_{DD} - V_{GS}}{R} = \frac{V_{DD} - \sqrt{\frac{2I_{D1}}{\beta_1}} - V_T}{R}$$

$$S_{V_{DD}}^{I_{REF}} > 1 \quad (\text{approaches 1 for large } V_{DD})$$



Note: This circuit does *not* address temperature and process dependence, both of which can have a significant impact on performance.

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MOS Widlar Current Reference

Operation

$$V_{GS1} - V_{GS2} - I_{OUT}R_2 = 0$$

$$I_{OUT}R_2 + V_{ON2} - V_{ON1} = 0$$

Assuming strong inversion and $\lambda \rightarrow 0$

$$I_{OUT}R_2 + \sqrt{\frac{2I_{OUT}}{k'(W_2/L_2)}} - V_{ON1} = 0$$

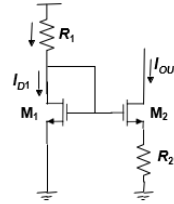
Solving for $\sqrt{I_{OUT}}$ gives

$$\sqrt{I_{OUT}} = \frac{-\sqrt{\frac{2}{k'(W_2/L_2)}} + \sqrt{\frac{2}{k'(W_2/L_2)} + 4R_2V_{ON1}}}{2R_2} \quad \text{where } V_{ON1} = \sqrt{\frac{2I_{REF}}{k'(W_1/L_1)}}$$

Differentiating I_{OUT} with respect to V_{DD} gives

$$\frac{1}{2\sqrt{I_{OUT}}} \frac{dI_{OUT}}{dV_{DD}} = \frac{1}{\sqrt{2/(k'(W_2/L_2) + 4R_2V_{ON1})}} \frac{dV_{ON1}}{dV_{DD}}, \quad \frac{dV_{ON1}}{dV_{DD}} = \frac{V_{ON1}}{2I_{REF}} \frac{dI_{REF}}{dV_{DD}}$$

$$\therefore S_{V_{DD}}^{I_{OUT}} = S_{V_{DD}}^{I_{REF}} = \frac{V_{ON1}}{\sqrt{V_{ON1}^2 + 4I_{OUT}R_2V_{ON1}}} S_{V_{DD}}^{I_{REF}} \approx \frac{V_{ON1}}{\sqrt{4V_{ON1}^2}} S_{V_{DD}}^{I_{REF}} = 0.5 S_{V_{DD}}^{I_{REF}}$$



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Example: MOS Widlar Current Reference

For the MOS Widlar current reference, find I_{OUT} if $I_{IN} = 100 \mu A$, $R_2 = 4 k\Omega$, $K' = 200 \mu A/V^2$, and $W_2/L_2 = W_1/L_1 = 25$. Assume that the temperature is $27^\circ C$ and that $\eta = 1.5$ also, find the sensitivity of I_{OUT} with respect to V_{DD}

$$V_{ON1} = \sqrt{\frac{2I_{IN}}{K'(W_1/L_1)}} = \sqrt{\frac{2 \cdot 100}{200 \cdot 25}} = 0.2V$$

$$\sqrt{I_{OUT}} = \frac{-\sqrt{\frac{2}{200 \cdot 25}} + \sqrt{\frac{2}{200 \cdot 25} + 4(0.004)0.2}}{20.004} \sqrt{\mu A} = 5\sqrt{\mu A} \Rightarrow I_{OUT} = 25\mu A$$

Notice that $V_{ON2} = V_{ON1} - I_{OUT}R_2 = 0.2 - (25)(0.004) = 0.1V > 2\eta V_t = 78mV$

so both transistors are in strong inversion.

For the sensitivity calculations, assume that $V_{DD} \gg V_{GS1}$. Therefore, $I_{IN} \approx V_{DD}/R_1$

$$S_{V_{DD}}^{I_{REF}} = \frac{V_{ON1}}{\sqrt{4V_{ON2}^2}} S_{V_{DD}}^{I_{IN}} \approx \frac{V_{ON1}}{\sqrt{4V_{ON2}^2}} = 0.5$$

Therefore, a 10% variation in V_{DD} causes a 5% variation in I_{OUT} .



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MOS Peaking Current Reference

Strong inversion operation

$$\begin{aligned} V_{GS1} &= I_{DN}R & V_{GS2} &= 0 \\ V_{DN2} &= V_{DN1} - I_{DN}R \\ I_{OUT} &= \frac{k'(W_2/L_2)}{2} V_{DN2}^2 = \frac{k'(W_2/L_2)}{2} (V_{DN1} - I_{DN}R)^2 \end{aligned}$$

where

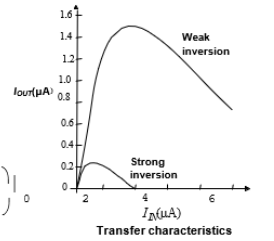
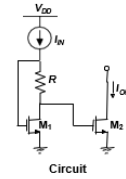
$$V_{DN1} = \sqrt{\frac{2I_{DN}}{K'(W_1/L_1)}}$$

Weak inversion operation

$$V_{GS2} - V_T \approx nV_T \ln\left(\frac{I_{DN}}{(W_1/L_1)I_T}\right) - I_{DN}R$$

If the transistors are identical and $V_{DS2} > 3V_T$

$$I_{OUT} = \frac{W_2}{L_2} I_T \exp\left(\frac{V_{GS2} - V_T}{nV_T}\right) \approx I_{DN} \exp\left(\frac{-I_{DN}R}{nV_T}\right)$$



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Threshold Referenced Current Reference

Operation

$$I_{OUT} = \frac{V_{GS1}}{R_2} = \frac{V_T + \sqrt{\frac{2I_{REF}}{K'(W_1/L_1)}}}{R_2} \approx \frac{V_T}{R_2} \text{ if } V_T > V_{ON1}$$

The sensitivity of I_{OUT} with respect to V_{DD} is

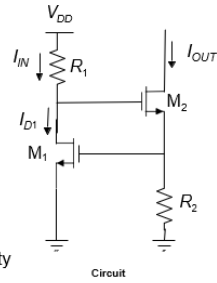
$$S_{V_{DD}}^{I_{REF}} = \left(\frac{V_{ON1}}{I_{OUT} R_2} \right) S_{V_{DD}}^{I_{REF}} = \left(\frac{V_{ON1}}{2V_{GS1}} \right) S_{V_{DD}}^{I_{REF}}$$

For example, if $V_T = 1V$, $V_{ON1} = 0.1V$ and the input sensitivity is approximately 1:

$$S_{V_{DD}}^{I_{REF}} = \left(\frac{0.1}{2 \cdot 1.1} \right) = 0.045$$

Therefore, if V_{DD} changes by 10%, I_{REF} or I_{OUT} changes by only 0.45%.

Note: This circuit has lower noise than the Widlar current source.



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Summary

- Simple current references
- Widlar current references
- Peaking current references
- Threshold-referenced references
- The concept of sensitivity



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Designing Bias Circuits

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Introduction: Designing Bias Circuits

- Both circuit design techniques (such as cascoding) and layout techniques (such as common centroid) to reduce errors due to mismatch
- The effects of both random noise and interference (crosstalk) on bias circuits, and the consequences of noise on the circuit being biased
- Circuit design and layout techniques to reduce the effects of noise and interference

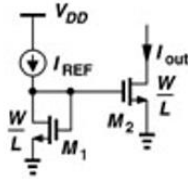


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Matching Considerations

Fundamental principle of integrated circuit design: Identical devices under identical conditions show identical behavior

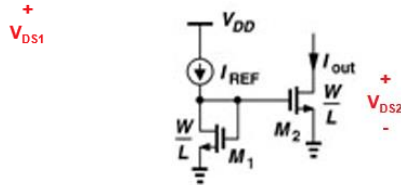
Example: In an ideal current mirror, drain currents I_{D1} and I_{D2} equal $I_{out} = I_{REF}$



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Matching Considerations: Reality

- Fundamental principle of integrated circuit design
 - Identical devices ... $M1 \neq M2$
 - ... under identical conditions ... $V_{DS1} \neq V_{DS2}$
 - ... show identical behavior $I_{D1} \neq I_{D2}$
- Example: real current mirror
- Drain currents I_{D1} , I_{D2} are unequal!
- Mismatch: $I_{out} \neq I_{REF}$



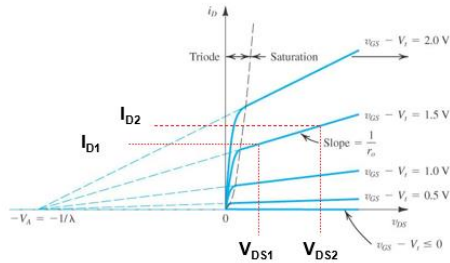
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Mismatch Causes: Systematic

Device V_{DS} unequal

- Channel length modulation

$$i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (v_{GS} - V_t)^2 (1 + \lambda v_{DS})$$



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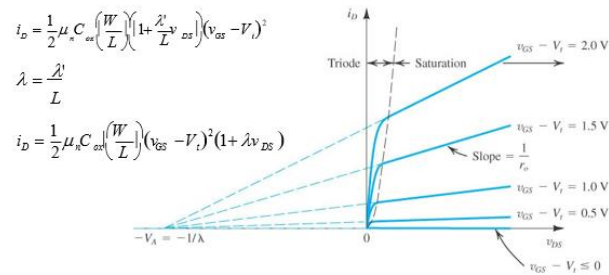
Systematic Solutions: Circuit Design

Device V_{DS} unequal

- Channel-length modulation

Longer L

- Reduces channel-length modulation parameter λ
- Disadvantage: Increases size, so you need to increase W by the same factor to keep the same voltage swing in saturation:



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Systematic Solutions: Circuit Design (continued)

Device V_{DS} unequal

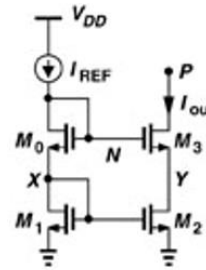
- Channel-length modulation

Longer L

- Reduces channel-length modulation parameter λ
- Disadvantage: Increases size, so you need to increase W by the same factor to keep the same voltage swing in saturation

Cascode

- Reduces variation in V_{DS}
- Disadvantage: Reduces available voltage swing at output node, so you need to increase sizes to maintain voltage range



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Systematic Solutions: Circuit Design (continued)

Device VDS unequal

- Channel-length modulation

Longer L

- Reduces channel-length modulation parameter λ
- Disadvantage: Increases size, so you need to increase W by the same factor to keep the same voltage swing in saturation

Cascode

- Reduces variation in VDS
- Disadvantage: Reduces available voltage swing at the output node, so you need to increase sizes to maintain the voltage range

Which approach results in less total area?

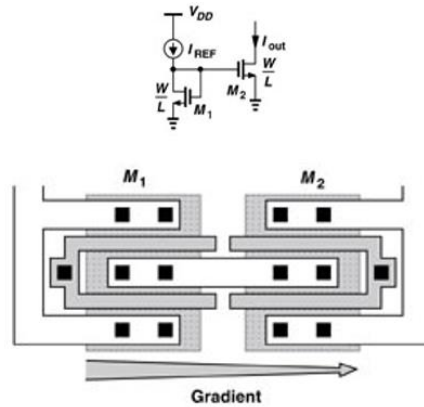
- Depends on details of individual design
- Other considerations (for example, capacitance) usually favor cascode



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Systematic Solutions: Layout

One source of M1-M2 parameter variation: process gradient

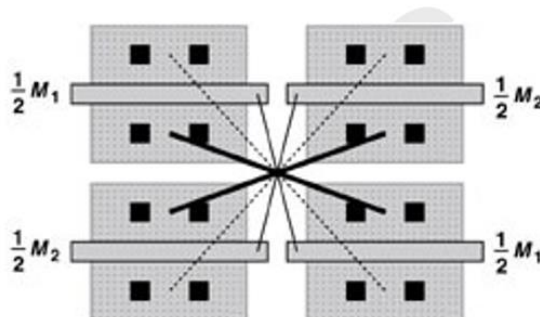


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Systematic Solutions: Common Centroid

Concept: Decompose each device into two halves

- Centroid in the same location: gradients cancel

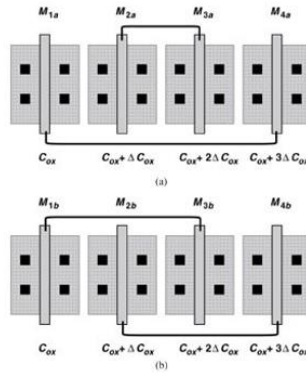


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Common Centroid in Practice

Concept: Decompose each device into two halves

- Centroid in the same location: gradients cancel



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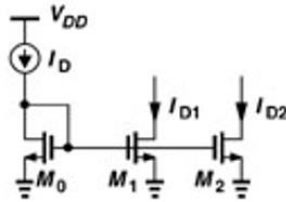
Mismatch Causes: Random

Matching between two current sources

- Random variations in W_1 and L_1 , and W_2 and L_2

Device parameters unequal

- Random variation in μ_{Cox} , threshold voltage V_{TH}



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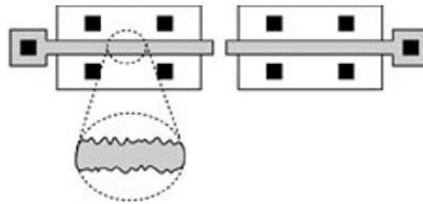
Mismatch Causes: Random (continued)

Device dimensions unequal

- Random variations in W and L

Device parameters unequal

- Random variation in μ_{Cox} , threshold voltage V_{TH}

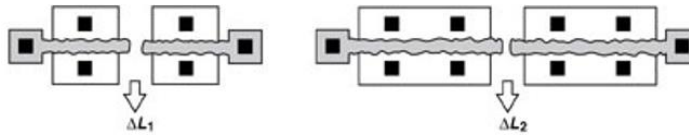


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Random Mismatch Causes: Solutions

Increase device area

- Random variations in W and L are a smaller fraction of the total dimension
- Also observed to reduce the mismatch in device parameters μ_{Cox} , threshold voltage V_{TH}



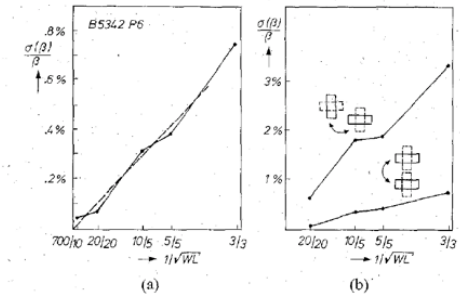
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Increasing Device Area

- Standard deviation of the mismatch in $\mu C_{ox} W/L$ (β) decreases as $\sqrt{\text{area}}$:

$$\Delta\left(\mu C_{ox} \frac{W}{L}\right) = \frac{A_K}{\sqrt{WL}}$$

- Parameter A_K is characteristic of the process



Standard deviation of (a) the current factor β versus the inverse transistor area for an NMOS transistor pair and (b) a comparison of the standard deviation of β for parallel and rotated placement.

Pelgrom, Duinmaier, and Welbers, "Matching Properties of MOS Transistors," IEEE Journal of Solid-State Circuits, pp. 1433-1439, Oct. 1989.

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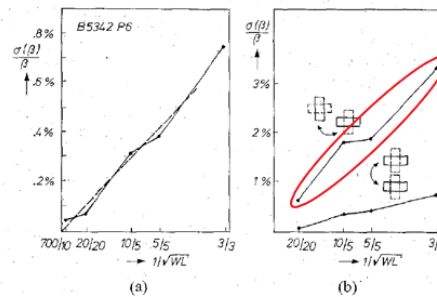
Increasing Device Area (continued)

- Standard deviation of the mismatch in $\mu C_{ox} W/L$ (β) decreases as $\sqrt{\text{area}}$:

$$\Delta\left(\mu C_{ox} \frac{W}{L}\right) = \frac{A_K}{\sqrt{WL}}$$

- Parameter A_K is characteristic of the process.

Note: Regarding the layout, the variation is greater if the devices are rotated.



Standard deviation of (a) the current factor β versus the inverse transistor area for an NMOS transistor pair and (b) a comparison of the standard deviation of β for parallel and rotated placement.

Pelgrom, Duinmaijer, and Welbers, "Matching Properties of MOS Transistors," IEEE Journal of Solid-State Circuits, pp. 1433-1439, Oct. 1989



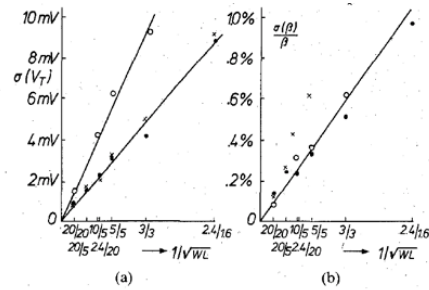
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Increasing Device Area (continued)

- Standard deviation of the mismatch in $\mu C_{ox} W/L$ (β) decreases as $\sqrt{\text{area}}$:

$$\Delta(V_{TH}) = \frac{A_{VTH}}{\sqrt{WL}}$$

- Parameter A_{VTH} is characteristic of the process
- A_{VTH} tends to scale down with t_{ox}
- A_{VTH} tends to scale down with minimum L



Standard deviation of (a) V_T and (b) β versus the square root of the inverse area of an NMOS transistor pair for a 25-nm gate-oxide process (dots) and a 50-nm gate-oxide process (circles). Crosses are measurements on devices processed in a 25-nm process with direct wafer writing. The cross for the 2.4/1.6 device is at 2.2 percent.

Reference: Pelgrom, Duinmaier, and Weibers, "Matching Properties of MOS Transistors," IEEE Journal of Solid-State Circuits, pp. 1433-1439, Oct. 1989

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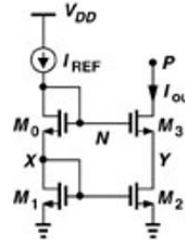
Testing Your Understanding: Matching Calculation

Analysis example: Given a cascode mirror (so V_{DS} mismatch is not a factor)

- Gate overdrive for M1, M2 is 0.3V ($V_{GS1} - V_{TH}$)
- M1, M2 are minimum length devices in 0.5 μm process with $A_{VTH} = 10 \text{ mV } \mu\text{m}$ and $A_K = 0.025 \text{ } \mu\text{m}$
- All devices $W/L = 10/0.5$

Determine

- Variation in I_D due to V_{TH}
- Variation in I_D due to $\mu C_{ox} W/L$
- I_D variation with both errors (assume independent)



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Variation Due to mCoxW/L (b)

Given a cascode mirror (so VDS mismatch is not a factor)

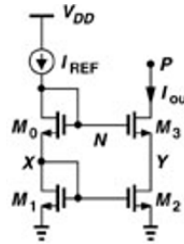
- M1, M2 are minimum length devices in 0.5mm process with $AK = 0.025 \text{ m}$
- All devices $W/L = 10/0.5$

Determine variation in mCoxW/L:

$$\sigma_{(\Delta\beta/\beta)} = \frac{A_K}{\sqrt{WL}} = \frac{0.025 \text{ m}}{\sqrt{(0.5\mu\text{m})(10\mu\text{m})}} = 1.1\%$$

Because I_D is proportional to mCoxW/L,
fractional variation in I_D :

$$(\Delta I_D / I_D) = 1.1\%$$



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Variation Due to V_{TH}

Given a cascode mirror (so V_{DS} mismatch is not a factor)

- Gate overdrive for M1, M2 is 0.3V ($V_{GS1} - V_{TH}$).
- $A_{V_{TH}} = 10 \text{ mV } \mu\text{m}$
- All devices $W/L = 10/0.5$

Determine variation in V_{TH} : $\sigma_{V_{TH}} = \frac{10 \text{ mV } \mu\text{m}}{\sqrt{(0.5 \mu\text{m})(10 \mu\text{m})}} = 4.5 \text{ mV}$

- Changes $V_{OV} = V_{GS} - V_{TH}$ by 4.5 mV.
- Sensitivity of I_D to overdrive V_{OV} in square law = 2:

$$S_{V_{OV}}^{I_D} = \frac{V_{OV}}{I_D} \frac{d}{dV_{OV}} \beta V_{OV}^2 = \frac{V_{OV}}{I_D} 2\beta V_{OV} = 2$$

- $4.5 \text{ mV} / 300 \text{ mV} = 1.5\%$ change in V_{OV} $(\Delta I_D / I_D) = 3\%$



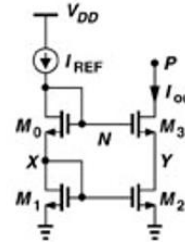
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Variation Due to V_{TH} (continued)

Determine ID variation with both errors

Assume independent – Standard deviations add in rss fashion:

$$\sigma_{(\Delta I_D/I_D)} = \sqrt{(0.03)^2 + (0.011)^2} = 3.2\%$$



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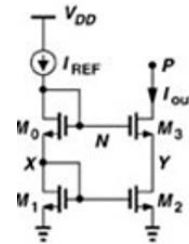
Testing Your Understanding: Design Example

Given a cascode mirror (so V_{DS} mismatch is not a factor)

- Gate overdrive for M1, M2 is 0.3V ($V_{GS1} - V_{TH}$)
- M2 are minimum length devices in 0.5 μm process with $A_{VTH} = 10 \text{ mV } \mu\text{m}$ and $A_K = 0.025 \mu\text{m}$
- Desired that standard deviation of variation in current mirror output is 2%

Determine

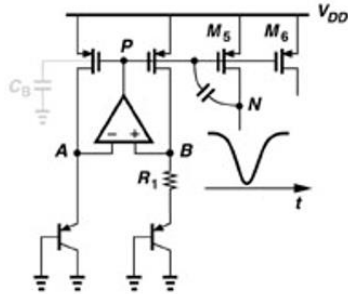
- Required W if all of variation was due to V_{TH}
- Required W if all of variation is due to $\mu C_{ox} W/L$
- W that meets 2% requirement with both errors (assume independent)



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Noise or Interference Affecting Circuit Operation

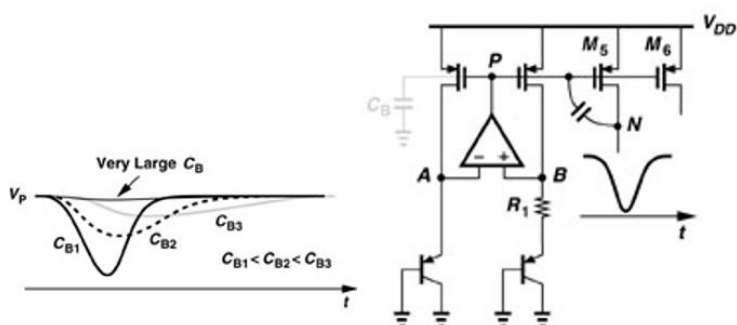
- Noise or interference affects bias point of circuit
- Noise affects output signal



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Solution: Reduce Impedance on Node P

Add bypass capacitor CB: Attenuates transient from N coupled to node P

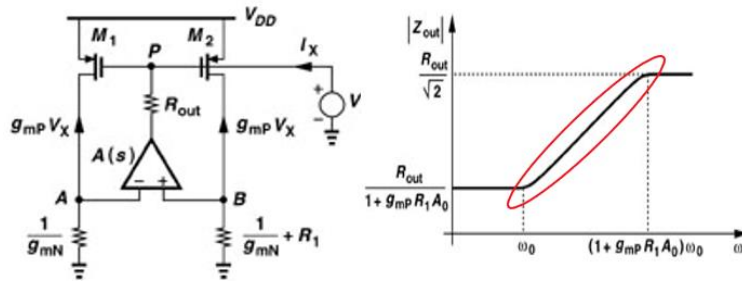


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Caution: Constraints on C_B

What is small-signal impedance at node P? Apply v_x , calculate i_x :

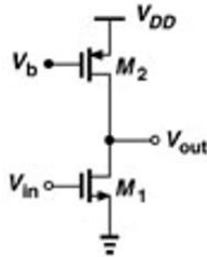
- Inductive over wide range of frequencies
- Can oscillate for some values of C_B
- Check stability over range of C_B values and process corners



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Noise Issues with Active Load

- Thermal noise of M_2 drain current
- Crosstalk coupling to V_b changing M_2 drain current



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Thermal Noise of Bias Current Affecting Signal

Input referred noise from drain current noise of

- M1 (gain MOSFET in common source configuration)
- M2 (active load MOSFET in current source configuration)

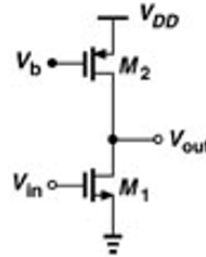
$$e_n^2 = \frac{8}{3} \frac{kT}{g_{m1}} + \frac{1}{g_{m1}^2} \frac{8}{3} kT g_{m2}$$

Rewrite in terms of M1 noise, increase from M2:

$$e_n^2 = \frac{8}{3} \frac{kT}{g_{m1}} \left(1 + \frac{g_{m2}}{g_{m1}} \right)$$

Result: You want $g_{m2} < g_{m1}$

- Difficulty: Same I_D ; requires $V_{OV2} > V_{OV1}$ reduces
- V_{OV} : Gate overdrive = $V_{GS} - V_{TH}$



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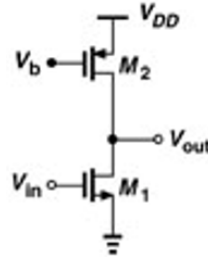
Crosstalk Coupling to V_b

Crosstalk coupling to V_b changes M2 drain current; couples to V_{out}

- Imagine V_{in} constant:
PMOS common source amplifier with V_b as input, M1 as load
- $\Delta V_{out} = g_{m2} (r_{o1} \parallel r_{o2}) \Delta V_b$

Solution strategy: Minimize terms in ΔV_{out} expression

- $(r_{o1} \parallel r_{o2})$
 - Cannot reduce this (reduces main amplifier gain)
- g_{m2}
 - Already reduced this (thermal noise)
- ΔV_b
 - Reduce coupling of interferers to V_b



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Summary

- Bias circuits can be affected by both systematic and random mismatch effects
- Solutions are available in both layout and circuit design
- Noise and interference (crosstalk) in bias circuits can degrade performance in signal path
- Solutions are available in both layout and circuit design techniques



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